



## FAMILY FACT FILE

### Key features

- Three main body parts: head, thorax (chest), and abdomen (belly and tail)
- A hard, protective external skeleton (exoskeleton)
- Three pairs of legs with joints
- Two antennae (feelers)
- Usually two pairs of wings

### When

The first insects appeared in the Devonian Period, 396 million years ago.

## Flies

- **When** 230 million years ago (Triassic) to now
- **Number of species today** About 240,000
- **Diet** Fly larvae (maggots) mostly eat rotting matter and flesh. Adults eat various liquid foods from nectar to blood.

Many insects can fly, but true flies are a particular family of insect that have two wings rather than four. Instead of a second pair of wings, they have two tiny knobs that flick back and forth to stabilize flight. Flies appeared at the same time as dinosaurs. Some of the early fly species almost certainly pestered the dinosaurs, sucking their blood and feeding on their eyes.



March fly

## Beetles

- **When** 260 million years ago (Late Permian) to now
- **Number of species today** Up to 1 million
- **Diet** Anything from pollen and nectar to fruit, flesh, other insects, rotting bodies, wood, and animal dung

Beetles evolved from flying insects that once had two pairs of wings. The front pair turned into hard, protective cases that folded

down over the rear wings to form a shield. The first flowering plants were probably pollinated by beetles. As flowering plants spread and evolved into new forms, so did the beetles. Today, there are so many species of beetle that they may well outnumber all other existing animal species combined.

Hydrophilus  
(water beetle)



▲ THE BEST insect fossils are found in amber – a hard, golden material formed of ancient tree resin. Tree resin is a thick, sticky liquid that oozes out of wounded trees, often trapping insects such as these ants.

## Cockroaches

- **When** 300–350 million years ago (Carboniferous Period) to now
- **Number of species today** Over 4,500
- **Diet** Rotten plant material

The first cockroaches looked much like those alive today. They scurried around on prehistoric forest floors, searching with their antennae (feelers) for dead plants. Termites evolved from wood-eating cockroaches that began living in colonies.

Archimylacris, a  
prehistoric cockroach

